

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE-AI	Date:	

Code of Conduct

I N Goutham/192472024(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N Goutham/192472024

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.
3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Q1. Reference Resolution Algorithm

Given Text

“Ravi met Arun at the library. He borrowed a book and later returned it.”

The task requires an algorithm to identify pronouns, link them to entities, rewrite the discourse and explain how context is used.

Pseudocode

ALGORITHM ReferenceResolution

INPUT: Discourse

 "Ravi met Arun at the library.

 He borrowed a book and later returned it."

OUTPUT: Resolved discourse

1. Extract all named entities from the discourse.
2. Store the entities in an entity list.
3. Identify all pronouns.
4. For each pronoun:
 - a. Find possible antecedents.
 - b. Check gender and number agreement.
 - c. Check semantic compatibility.
 - d. Apply discourse context and recency.
 - e. Select the most suitable antecedent.
5. Replace each pronoun with its selected antecedent.
6. Return the resolved discourse.

END ALGORITHM

Applying the Algorithm

Entities:

Ravi

Arun

book

Pronouns:

He

it

For **“He”**, both Ravi and Arun are grammatically possible because both are male. The sentence does not provide enough information to determine the antecedent with complete certainty.

A reasonable discourse-based interpretation is:

Ravi met Arun at the library. Ravi borrowed a book and later returned the book.

For **“it”**, the antecedent is clearly book, because **“it”** refers to the object borrowed and returned.

Resolved Discourse

Ravi met Arun at the library. Ravi borrowed a book and later returned the book.

Context-Based Validation

The algorithm uses:

- Entity identification
- Pronoun detection
- Gender/number agreement
- Semantic compatibility
- Recency
- Discourse context

The pronoun **“it”** is confidently resolved to **“book”** because the object is semantically compatible with both **“borrowed”** and **“returned.”** However, the antecedent of **“He”** is ambiguous from the supplied text alone; therefore, the selected resolution should be treated as a reasonable assumption rather than a guaranteed fact.

Q2. Text Coherence and Discourse Structure Algorithm

Given Discourse

“The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online.”

The task requires identification of cause-effect, sequence and discourse relations.

Pseudocode

ALGORITHM CoherenceAnalysis

INPUT: Set of sentences

OUTPUT: Discourse relations and structure

1. Split the text into individual sentences.
2. Identify temporal and causal indicators.
3. For each pair of adjacent sentences:
 - a. Determine whether a Cause-Effect relation exists.
 - b. Determine whether a Sequence relation exists.
 - c. Determine whether an Elaboration relation exists.
4. Construct a discourse tree.
5. Check whether each sentence logically connects
with the previous discourse.
6. Return the discourse structure.

END ALGORITHM

Applying the Algorithm

Sentence 1:

The roads were flooded after heavy rainfall.

This establishes the **cause/context**.

Sentence 2:

Therefore, schools were closed for the day.

The word “**Therefore**” explicitly indicates a **Cause** → **Effect** relationship.

Heavy rainfall



Roads flooded



Schools closed

Sentence 3:

Students attended classes online.

This represents the **resulting action/sequence** following school closure.

Discourse Structure

MAIN DISCOURSE



Cause → Effect



Roads flooded Schools closed



Heavy rainfall ↓

Students attended

classes online

Relations

Sentences	Relation	Explanation
S1 → S2	Cause-Effect	Flooded roads caused school closure
S2 → S3	Result/Sequence	Online classes followed school closure

Coherence Explanation

The text is coherent because each sentence logically follows from the previous sentence:

Heavy rainfall



Road flooding



School closure



Online classes

This creates a clear chain of events and allows the reader to understand the situation without additional information.

Q3. Dialogue Act Recognition Algorithm

Conversation

User: “Can you book a train ticket for me?”

Agent: “Sure, where would you like to travel?”

User: “I want to go to Chennai.”

Agent: “Your ticket has been booked.”

The assessment asks for classification as Request, Question, Inform, Confirmation or Action and requires a dialogue-act sequence.

Pseudocode

ALGORITHM DialogueActRecognition

INPUT: User-Agent conversation

OUTPUT: Dialogue act for each utterance

1. Read each utterance.
2. Identify keywords, sentence structure and intent.
3. If the user asks the agent to perform an action:
 classify as REQUEST.
4. If the speaker asks for information:
 classify as QUESTION.
5. If the speaker provides information:
 classify as INFORM.
6. If the system confirms completion:
 classify as CONFIRMATION.

7. If an actual system operation is performed:

classify as ACTION.

8. Store all identified dialogue acts in sequence.

9. Return dialogue-act sequence.

END ALGORITHM

Classification

Speaker	Utterance	Dialogue Act
User	Can you book a train ticket for me?	Request
Agent	Sure, where would you like to travel?	Question
User	I want to go to Chennai.	Inform
Agent	Your ticket has been booked.	Confirmation

Dialogue Act Sequence

REQUEST

↓

QUESTION

↓

INFORM

↓

CONFIRMATION

Evaluation

The sequence allows the conversational agent to understand the user's goal step by step:

1. **Request** identifies the user's intention to book a ticket.
2. **Question** collects missing information.
3. **Inform** supplies the destination.
4. **Confirmation** indicates successful completion.

Thus, dialogue acts help the system maintain context and respond appropriately.

Q4. Language Generation and Surface Realization

Semantic Input

Action: Buy

Agent: Student

Object: Book

Tense: Past

The assessment asks the algorithm to perform lexical selection, sentence structuring and surface realization.

Pseudocode

ALGORITHM SurfaceRealization

INPUT:

Action = Buy

Agent = Student

Object = Book

Tense = Past

OUTPUT:

Grammatically correct sentence

1. Select lexical items:

Agent → "student"

Action → "buy"

Object → "book"

2. Convert action into the required tense:

buy → bought

3. Construct sentence structure:

Subject + Verb + Object

4. Insert lexical items:

Student + bought + a book

5. Apply grammatical rules:

- Capitalize the first word.

- Add article "a" before singular count noun.
- Add period.

6. Return final sentence.

END ALGORITHM

Final Sentence

The student bought a book.

Lexical Selection

Student → student

Buy → bought

Book → book

Sentence Structure

Subject Verb Object

| | |

The student bought a book

Surface Realization

The semantic representation is converted into a natural sentence by:

1. Selecting suitable words.
2. Converting **buy** to its past form **bought**.
3. Arranging the words as **Subject + Verb + Object**.
4. Adding the appropriate article.
5. Applying capitalization and punctuation.

Grammatical Validation

The generated sentence is grammatically correct because:

- Subject = “The student”
- Verb = “bought”
- Object = “a book”
- Verb agrees with the past-tense requirement.

Q5. Interlingua + Statistical Machine Translation

Source Sentence

“The boy is playing football.”

The task requires source analysis, Interlingua representation, candidate target translations, statistical scoring and final translation.

Pseudocode

ALGORITHM HybridMachineTranslation

INPUT:

Source sentence = "The boy is playing football."

OUTPUT:

Final target-language translation

1. Analyze the source sentence.

2. Identify:

Subject = boy

Action = playing

Object = football

Tense = present continuous

3. Construct Interlingua:

PLAYING(boy, football)

TENSE = PRESENT_CONTINUOUS

4. Generate multiple candidate translations.

5. For each candidate:

a. Calculate translation score.

b. Check grammatical correctness.

c. Check semantic similarity.

d. Check fluency.

6. Select the candidate with the highest score.

7. Produce final translation.

END ALGORITHM

Step 1 — Source Analysis

Element	Value
Subject	Boy
Verb	Playing
Object	Football
Tense	Present Continuous
Meaning	A boy is currently playing football

Step 2 — Interlingua Representation

A simple Interlingua representation is:

AGENT = BOY

ACTION = PLAY

OBJECT = FOOTBALL

TENSE = PRESENT_CONTINUOUS

This representation captures the meaning independently of any particular language.

Step 4 — Statistical Scoring

A simplified scoring table can be used:

Candidate	Semantic Accuracy	Grammar	Fluency	Overall
Candidate 1	High	High	High	High
Candidate 2	High	High	Medium/High	High

Evaluation

The Interlingua representation helps preserve the underlying meaning before translation into the target language. It separates semantic interpretation from language-specific word ordering.

The statistical scoring approach helps compare possible translations based on factors such as fluency, grammatical correctness and semantic similarity.

Therefore, combining both approaches can:

- Reduce ambiguity.
- Preserve meaning.

- Improve grammatical quality.
- Select the most natural candidate.
- Produce more coherent translations.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudo code Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
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Date: _____	Date: _____	Date: _____